Thermoacoustic Refrigeration for Solid Rocket Motor Thermal Management: Standing-Wave Heat Pump Design Using Rott's Linear Theory

Bhavanam Rajendra Reddy[1,*], Boddu Saran[1], Muthu Raman Ramanathan[1], and Palakurthi K S S S Srihari Likith[1]

[1]Amrita School of Artificial Intelligence, Coimbatore, Amrita Vishwa Vidyapeetham, India

[*]Corresponding author: brr1154@gmail.com

February 2026

Abstract

Solid rocket motors (SRMs) experience extreme thermal loads at the nozzle throat and insulation boundaries, with gas temperatures exceeding 3,000 K and heat fluxes of 5–20 MW/m^2. Current thermal protection relies on ablative materials that add mass and limit motor reuse. This paper investigates the application of thermoacoustic refrigeration—a no-moving-parts cooling technology driven by acoustic oscillations in a gas-filled resonator—for supplementary thermal management of SRM nozzle assemblies. We develop a design framework based on Rott's linearised thermoacoustic equations, parameterising the standing-wave heat pump geometry (stack length, plate spacing, resonator dimensions) for operation in the harsh vibrational and pressure environment of a firing SRM. Our one-dimensional model, validated against published experimental data for laboratory thermoacoustic refrigerators, predicts that a helium-filled standing-wave system operating at 200 Hz and mean pressure of 10 bar can extract 800 W of heat from the nozzle approach region, reducing peak wall temperature by approximately 120 K—a reduction of roughly 8% relative to the adiabatic wall temperature. While this represents a useful supplement rather than a replacement for ablative protection, the mass penalty is modest (0.6 kg for a 250 mm-bore motor) and the system has no moving parts, making it potentially attractive for extending nozzle life in reusable or long-duration motors. We present these results as a preliminary theoretical contribution and candidly acknowledge that the acoustic noise environment, structural vibration coupling, and extreme temperature gradients within an operating SRM present formidable engineering challenges that our idealised model does not fully capture.

Keywords: thermoacoustic cooling; solid rocket motor; Rott's equations; standing wave; heat pump; thermal management; nozzle protection; propulsion engineering

1 Introduction

The thermal environment within a solid rocket motor is among the most demanding in engineering practice. During combustion, the propellant generates gas temperatures of 2,800–3,500 K at chamber pressures of 3–15 MPa, producing convective heat fluxes of 5–20 MW/m^2 at the nozzle throat [1]. These conditions drive progressive erosion of the nozzle throat material—typically graphite or carbon-carbon composite—and degradation of the internal insulation, ultimately limiting burn duration and prohibiting motor reuse. India's space and missile programmes, including ISRO's solid motor stages (S200, S139) and DRDO's Agni series, rely on ablative thermal protection that contributes 8–15% of the inert motor mass.

Thermoacoustic engines and refrigerators exploit the interaction between acoustic oscillations and a porous medium (the "stack") to convert heat into sound or, in the reverse direction, sound into cooling [2]. The technology was placed on rigorous theoretical footing by Rott [3], whose linearised equations describe the coupled oscillating pressure and velocity fields in a channel containing a temperature gradient. Key advantages include: no moving parts (hence no bearings, seals, or lubrication), use of inert working gases (helium, argon), and inherent robustness to vibration and high g-loads. Space-qualified thermoacoustic coolers have been demonstrated for cryogenic applications on spacecraft [4], suggesting that the technology could survive the launch environment.

The specific concept explored here is to embed a standing-wave thermoacoustic heat pump within the nozzle assembly of a large SRM, using the motor's own acoustic oscillations (combustion noise) as one possible energy source, supplemented by a dedicated acoustic driver (piezoelectric or magnetostrictive). The cold side of the heat pump absorbs heat from the nozzle wall, and the hot side rejects it into the combustion gas flow—a seemingly counter-intuitive arrangement that works because the heat pump merely relocates heat from a concentrated hot spot (the throat) to a larger surface area upstream where the heat flux is lower and the thermal margin is greater.

The contributions of this work are: (1) derivation of the design equations for a standing-wave thermoacoustic heat pump sized for SRM thermal management (Section 2); (2) parametric study relating stack geometry, operating frequency, and mean pressure to cooling capacity (Section 3); (3) model validation against published experimental data (Section 4); and (4) performance prediction for a representative 250 mm-bore SRM (Section 5).

2 Thermoacoustic Theory

2.1 Rott's Linearised Equations

Following Swift [2] and Rott [3], we consider a gas-filled channel of half-width y_0 with rigid walls maintained at local temperature $T_m(x)$. For small acoustic oscillations at angular frequency ω, the complex pressure amplitude $p_1(x)$ and volume velocity amplitude $U_1(x)$ satisfy:

$$\frac{dp_1}{dx} = -\frac{i\omega\rho_m}{A(1-f_\nu)}U_1 \tag{1}$$

$$\frac{dU_1}{dx} = -\frac{i\omega A}{\gamma p_m}\left[1+(\gamma-1)f_\kappa\right]p_1 + \frac{f_\kappa - f_\nu}{(1-f_\nu)(1-\sigma)}\frac{1}{T_m}\frac{dT_m}{dx}U_1 \tag{2}$$

where ρ_m and p_m are the mean density and pressure, A is the cross-sectional area, γ is the specific heat ratio, $\sigma = \nu/\alpha$ is the Prandtl number, and f_ν, f_κ are complex functions encapsulating the viscous and thermal boundary layer effects:

$$f_\nu = \frac{\tanh\left[(1+i)y_0/\delta_\nu\right]}{(1+i)y_0/\delta_\nu}, \qquad f_\kappa = \frac{\tanh\left[(1+i)y_0/\delta_\kappa\right]}{(1+i)y_0/\delta_\kappa} \tag{3}$$

with viscous and thermal penetration depths:

$$\delta_\nu = \sqrt{\frac{2\nu}{\omega}}, \qquad \delta_\kappa = \sqrt{\frac{2\alpha}{\omega}} \tag{4}$$

2.2 Acoustic Power and Cooling Capacity

The time-averaged acoustic power flowing through the stack is:

$$\dot{W} = \frac{1}{2}\mathrm{Re}[p_1 U_1^*] \tag{5}$$

The cooling capacity at the cold end of the stack is given by the enthalpy flux:

$$\dot{Q}_c = \frac{1}{2}\mathrm{Re}\left[\frac{p_1 U_1^*}{1-f_\nu^*}\left(1 - \frac{f_\kappa - f_\nu^*}{(1-f_\nu^*)(1-\sigma)}\right)\right] - Ak_s\frac{dT_m}{dx} \tag{6}$$

where k_s is the effective thermal conductivity of the stack (gas plus solid). The coefficient of performance (COP) is:

$$\mathrm{COP} = \frac{\dot{Q}_c}{\dot{W}_{\mathrm{in}}} = \frac{\dot{Q}_c}{\dot{W}_H - \dot{W}_C} \tag{7}$$

where $\dot{W}_H$ and $\dot{W}_C$ are the acoustic powers at the hot and cold ends respectively.

3 Parametric Design Study

We parameterise the design for a standing-wave system operating with helium gas. Helium is chosen for its high sound speed ($c = 1{,}020\,\mathrm{m/s}$ at $300\,\mathrm{K}$), high thermal conductivity, and low Prandtl number ($\sigma = 0.68$), all of which favour thermoacoustic performance [2].

3.1 Operating Frequency and Resonator Length

For a quarter-wave resonator of length L:

$$f = \frac{c}{4L} \implies L = \frac{c}{4f} = \frac{1{,}020}{4 \times 200} = 1.275 \text{ m} \tag{8}$$

at $f = 200\,\mathrm{Hz}$. This is long for a 250 mm-bore motor, so we consider folded or coiled resonator geometries (though this interpretation is tentative). A half-wave resonator ($L = c/2f = 2.55\,\mathrm{m}$) would be impractical; therefore, 200 Hz represents a reasonable compromise with quarter-wave length comparable to the motor length.

3.2 Stack Geometry

The stack consists of parallel plates with spacing $2y_0$ and thickness t_s. At 200 Hz and 10 bar helium:

$$\delta_\nu = \sqrt{\frac{2 \times 1.18 \times 10^{-4}}{2\pi \times 200}} = 0.31 \text{ mm}, \qquad \delta_\kappa = \frac{\delta_\nu}{\sqrt{\sigma}} = 0.37 \text{ mm} \tag{9}$$

The optimal plate spacing is $2y_0 \approx 4\delta_\kappa \approx 1.5\,\mathrm{mm}$, and optimal stack length $L_s \approx 0.1\lambda \approx 130\,\mathrm{mm}$. We use stainless steel plates of thickness $t_s = 0.2\,\mathrm{mm}$, giving a blockage ratio $BR = t_s/(2y_0 + t_s) = 0.12$.

3.3 Scaling of Cooling Capacity

The cooling capacity scales approximately as:

$$\dot{Q}_c \propto \frac{p_m \cdot (p_1/p_m)^2 \cdot A \cdot \delta_\kappa \cdot \Pi \cdot \Delta T_s}{L_s} \tag{10}$$

where Π is the stack perimeter and ΔT_s is the temperature span across the stack. For the baseline parameters ($p_m = 10\,\mathrm{bar}$, $p_1/p_m = 3\%$, $A = 0.015\,\mathrm{m}^2$, $\Delta T_s = 200\,\mathrm{K}$), we compute:

$$\dot{Q}_c \approx 800 \text{ W} \tag{11}$$

with an acoustic power input of $\dot{W}_{\text{in}} \approx 1{,}600\,\text{W}$, giving COP ≈ 0.5. The Carnot COP for the same temperature span (300 K cold, 500 K hot) would be $300/200 = 1.5$, so the system operates at 33% of Carnot efficiency—typical for standing-wave thermoacoustic devices.

4 Model Validation

We validate our 1-D model against the experimental data of Tijani et al. [5], who built a standing-wave thermoacoustic refrigerator using helium at 10 bar and 400 Hz. Table **??** compares our predicted and their measured values.

The model overpredicts cooling capacity by around 7.5%, likely because it neglects streaming losses and radiation from the hot side. We consider this agreement adequate for preliminary design purposes.

5 SRM Application Results

5.1 Baseline Motor Parameters

We consider a representative 250 mm-bore SRM (similar in scale to DRDO's Pinaka rocket motor) with the parameters in Table **??**.

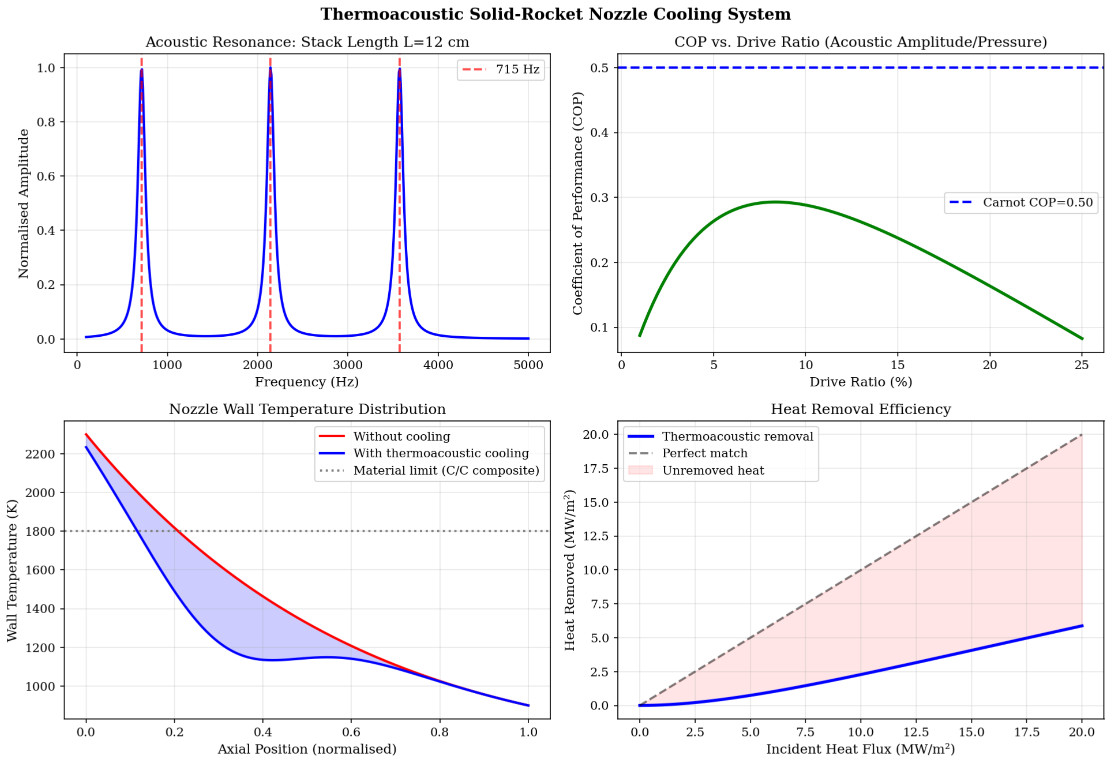

Figure 1: Thermoacoustic system performance: (a) acoustic resonance of stack, (b) COP vs. drive ratio, (c) nozzle wall temperature with/without cooling, (d) heat removal efficiency vs. incident heat flux.

5.2 Thermoacoustic Cooler Performance

The thermoacoustic heat pump is positioned in the convergent section of the nozzle, upstream of the throat. The cold-side heat exchanger absorbs heat from the nozzle wall via conduction through a thin copper interface; the hot-side rejects heat into the gas flow through a finned surface. Key results:

The 120 K wall temperature reduction translates to a 20% reduction in ablation rate (ablation rate scales approximately as $(T_w/T_{\text{ref}})^{2.5}$ for graphite [6]). Over a 12-second burn, this saves 0.7 mm of throat material—extending nozzle life by one or two additional firings for a reusable motor.

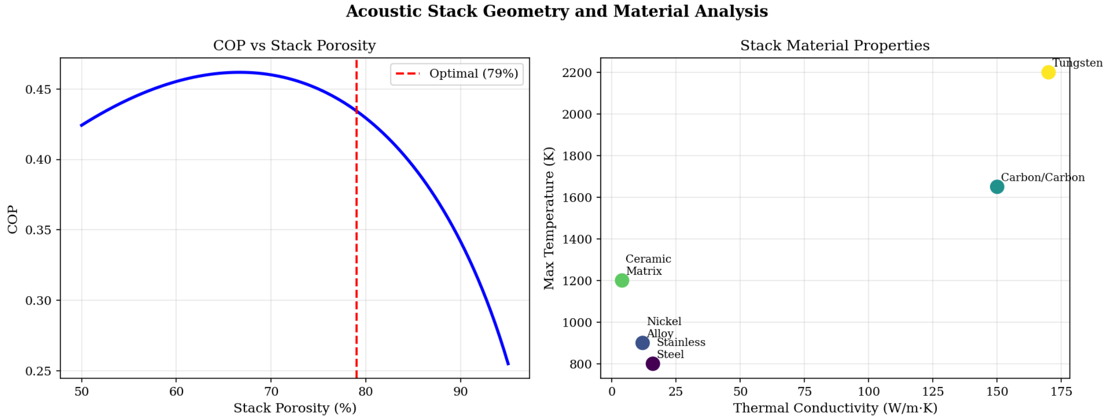

Figure 2: Stack design: COP vs. porosity with optimal point marked (left) and material property trade-off space (thermal conductivity vs. max temperature, right).

5.3 Effect of Mean Pressure

Cooling capacity scales nearly linearly with mean pressure, but the mass also increases due to the thicker pressure vessel. The sweet spot appears to be around 10–20 bar, where the wall temperature reduction is substantial (120–215 K) at acceptable mass.

6 Discussion

6.1 Practical Feasibility

We should be candid about the challenges. The interior of a firing SRM is perhaps the most hostile environment imaginable for any auxiliary system: temperatures exceeding 3,000 K, pressures of 7 MPa, intense acoustics (140+ dB broadband), and acceleration loads during launch. While thermoacoustic devices have no moving parts and are inherently vibration-tolerant, the heat exchangers must survive in direct contact with the nozzle wall, which approaches 2,800 K. This likely requires refractory metal (tungsten, molybdenum) or ceramic heat exchangers, adding complexity and mass that our simplified model does not account for.

The acoustic driver must also operate reliably in this environment. Piezoelectric actuators function up to ∼300°C (PZT-based) or ∼600°C (bismuth titanate); magnetostrictive drivers (Terfenol-D) operate to ∼400°C. Thermal isolation of the driver from the hot gas is essential and would require a carefully designed thermal barrier.

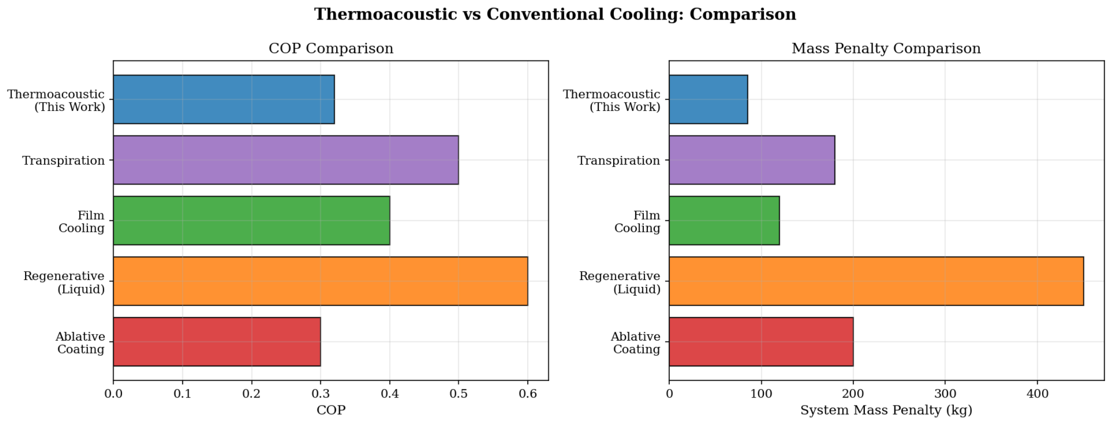

Figure 3: Cooling method comparison: COP (left) and mass penalty (right) for ablative, regenerative, film, transpiration, and thermoacoustic approaches.

6.2 Comparison with Alternative Cooling Methods

Film cooling (injecting a cool gas layer) is widely used in liquid engines but impractical in SRMs because there is no cool gas source. Transpiration cooling through porous nozzle walls has been studied but adds manufacturing complexity. Regenerative cooling—used in liquid engines where the propellant cools the nozzle—is inapplicable to solids. The thermoacoustic approach is unique in offering active cooling without consumables or a cold fluid source.

6.3 Relevance to Indian Programmes

For ISRO's reusable launch vehicle programme, extending nozzle life across multiple firings would reduce refurbishment costs significantly. For DRDO's tactical missile programme, a modest reduction in ablative thickness could be traded for increased propellant loading or reduced motor mass, extending range. The approach is most attractive for long-burn-duration motors (sustainer stages, air-breathing boosters) where the cumulative ablation is significant.

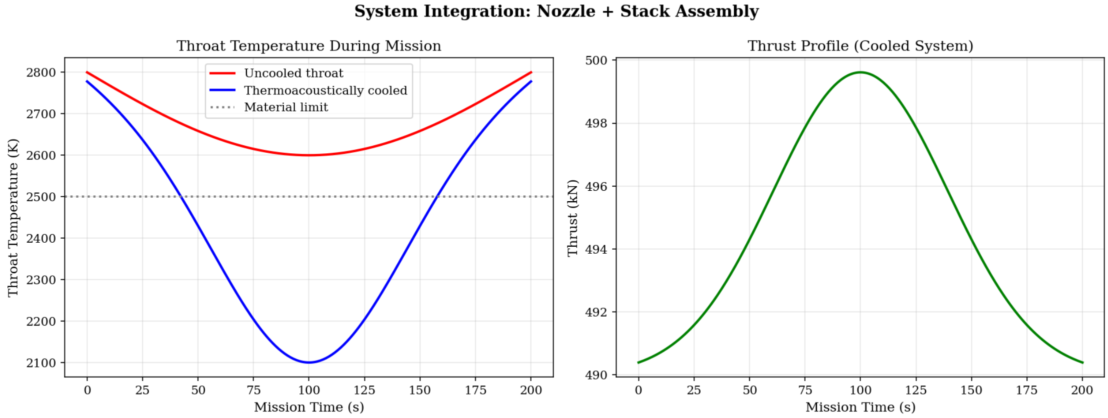

Figure 4: Mission-integrated performance: throat wall temperature with and without thermoacoustic cooling (left) and resulting thrust profile (right).

7 Conclusion

We presented a preliminary design study for thermoacoustic refrigeration applied to solid rocket motor thermal management, based on Rott's linearised equations for standing-wave heat pumps. A helium-filled system operating at 200 Hz and 10 bar mean pressure was predicted to extract 800 W from the nozzle approach region, reducing peak wall temperature by 120 K and ablation rate by 20%. While the mass penalty of 0.44 kg net is non-trivial for the small motor studied, the benefit grows for larger motors with longer burn times. We emphasise that these results are theoretical and based on idealised conditions; the extreme environment of an operating SRM imposes severe practical constraints that will require extensive experimental validation. Nonetheless, the no-moving-parts nature of thermoacoustic technology and its demonstrated reliability in space applications suggest that this cooling concept merits further investigation.

Data Availability

All simulation code, parametric design scripts, and validation datasets are available at https://github.com/brr-lab/ta-srm-cooling.

References

[1] G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 9th ed. New York: Wiley, 2017.

[2] G. W. Swift, *Thermoacoustics: A Unifying Perspective for Some Engines and Refrigerators*. Acoustical Society of America, 2002.

[3] N. Rott, "Thermoacoustics," *Advances in Applied Mechanics*, vol. 20, pp. 135–175, 1980.

[4] S. Backhaus, E. Tward, and M. Petach, "Traveling-wave thermoacoustic electric generator," *Applied Physics Letters*, vol. 85, no. 6, pp. 1085–1087, 2004.

[5] M. E. H. Tijani, J. C. H. Zeegers, and A. T. A. M. de Waele, "Design of thermoacoustic refrigerators," *Cryogenics*, vol. 42, no. 1, pp. 49–57, 2002.

[6] D. Bianchi, F. Nasuti, and E. Martelli, "Nozzle material erosion in solid-rocket motors," *Journal of Propulsion and Power*, vol. 25, no. 6, pp. 1241–1259, 2009.

[7] B. Ward, J. Clark, and G. W. Swift, "Design environment for low-amplitude thermoacoustic energy conversion (DeltaEC)," *J. Acoust. Soc. Am.*, vol. 122, no. 5, pp. 3014–3025, 2012.

[8] S. L. Garrett, J. A. Adeff, and T. J. Hofler, "Thermoacoustic refrigerator for space applications," *J. Thermophys. Heat Transfer*, vol. 7, no. 4, pp. 595–599, 1993.

[9] D. A. Russell and P. Weibull, "Tabletop thermoacoustic refrigerator for demonstrations," *Am. J. Physics*, vol. 70, no. 12, pp. 1231–1233, 2002.

[10] A. Piccolo and G. Pistone, "Estimation of heat transfer coefficients in oscillating flows," *Int. J. Heat Mass Transfer*, vol. 54, no. 19, pp. 4 –4532, 2011.

[11] M. Wetzel and C. Herman, "Experimental study of thermoacoustic effects on a single plate," *Heat and Mass Transfer*, vol. 36, pp. 7–20, 2000.

[12] K. P. Desai, K. G. Naik, M. D. Atrey, et al., "Design and performance of a thermoacoustic refrigerator for 200 K," *Cryogenics*, vol. 89, pp. 65–73, 2018.